Effect of experimental habitat connectivity on dung beetle communities

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Abstract

Habitat fragmentation threatens biodiversity across the globe as habitat loss, isolation, and edge effects become increasingly prevalent. Corridors have become an important tool in order to combat the negative effects of fragmentation, however they are difficult to study in natural systems without incurring confounding effects. To observe changes in insect community composition as an effect of landscape features we sampled dung beetles in a landscape scale experiment. We did not see a difference in species richness or diversity, but dung beetle abundances were higher in continuous forest habitat and open habitat patches connected by a corridors than in isolated patches.

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TO DO

Introduction

1. any details missing? citations all ok or do we need other/better ones?

Methods/Results

1. Do we need to do more with functional group? Which are most common, biomass by functional group, effect of corridor on functional group, etc.?

Discussion

1. Anything missing? any topics need to be considered in more depth? What could we make more efficient?

Figures & Tables

1. Make more purty

Supplement

1. table of individual biomass/total biomass of each species

INTRODUCTION

Loss of biological diversity is a leading threat to the sustainability of the biosphere (1) and is largely caused by habitat loss and fragmentation (2). Landscape corridors (strips of habitat connecting otherwise isolated habitat patches) are hypothesized to reduce the negative effects of fragmentation by facilitating gene flow and the movement of organisms, thereby preventing local extinctions and increasing species diversity (3). Corridors have become a central feature of ecological management plans worldwide, but evidence of their effectiveness has lagged behind the push for their implementation (4). Although a number of

experimental studies have demonstrated positive corridor effects on single species (5–7), few have examined corridor effects on entire communities. All of these studies were conducted at very small spatial scales (10 cm2 to 10 m2), and taken together they have yielded equivocal results (8–11).

Intact habitats continue to be fragmented by human activities (Taubert et al. 2018, Díaz et al. 2019, Ma et al. 2023). The drivers of habitat fragmentation are complex and multifaceted, and there continues to be a debate about whether its effects on biodiversity are primarily positive or negative (Fahrig 2003, Fletcher et al. 2018). Among the best documented negative effects are the local extinction of species from habitat patches, which can result in lower-patch level richness and diversity. Habitat corridors are thought to be one of rth ameliorate the negative effexcts f

However, there is little doubt that as habitats are broken down community structures are significantly altered (Harrison and Bruna 1999, Haddad et al. 2003, Jennings and Tallamy 2006, Laurance et al. 2018). This alteration of structure typically lends to loss in biodiversity on a global scale and interruptions in ecosystem processes and functions (Haddad 2015).

Habitat corridors are thought to be an important tool for ameliorating the negative consequences of fragmentation (Haddad et al. 2003). If they can facilitate the dispersal of individuals between patches, they can increase the likley hood of local and landscape population viability (Forman 1995). The resulting changes in species composition are important to identify because any species impacted would have corresponding effects depending on how they interact with the ecosystem (Zhou et al. 2023). Any gain or loss in key members of an community could disrupt processes which on their own could shape ecosystems (Cuke and Srivastava 2016), or effect other organisms which rely on said interaction (Wu et al. 2011). Because of this dynamic it becomes necessary to understand responses by species compositions at all taxonomic levels and potential trophic cascades

resulting from changes in habitat structure and connectivity (Debinski and Holt 2000).

By measuring changes in biodiversity and species richness within experimental designs we were able to isolate factors might be contributing to ecological patterns and processes (Resasco et al. 2017, Fletcher Jr. et al. 2023). Past studies have endeavored to experimentally measure changes in community compositions as a result of connecting habitats with corridors (Tewksbury et al. 2002, Collins et al. 2017, Graham et al. 2022) Yet very few have directly compared matrix and patch populations. Land use is different from one species to another so it is vital to understand where compositions are distributed and what processes might be driving population differences(Haddad 1999).

Dung beetles have emerged as a model system with which to test spatial ecology hypotheses (Roslin 2000, Rös et al. 2012).They are an incredibly well studied group of insects which are well known for driving a multitude of ecosystem functions (Hasan et al. 2024). The removal, breakdown, and burial of animal feces drive important ecosystem interactions provided by dung beetles enhancing nutrient cycling and soil quality, the reduction of breeding sites for parasites, and a reduction in methane emissions from dung (Nichols et al. 2008, Iwasa et al. 2015, Slade et al. 2016b). Local assemblages of dung beetles can be species-rich with species comprising a broad range of functional traits [e.g., size, foraging style, resource-use, (Ospina-Garcés et al. 2018, deCastro-Arrazola et al. 2023)]. Previous studies have shown that isolated patches of habitat frequently have lower dung beetle diversity and abundance than areas of continuous habitat, as well as documented their presence in linear strips of habitat that resemble corridors (Gray et al. 2022). Past studies have also focused on how landscape structure alters the community compositions of dung beetles(Costa et al. 2017), yet large landscape scale experimental studies with carefully controlled and replicated treatments are non-existent for this model species.

Here, we aim to gain an understanding of how dung beetles, a group of insects well known for strong dispersal ability in order to compete for ephemeral resources (Hanski and

Cambefort 1991), interact with corridors in their landscapes. We sampled dung beetle communities in experimental landscapes developed for the express purposes of comparing connected and isolated patches, as well as the effects of patch area to edge ratio and distance to edge (Tewksbury et al. 2002). We do so to address these questions:

(1) How does the abundance and biomass of dung beetles differ among and between isolated and connected patches and how does this compare with abundance in matrix habitat?
(2) Does species richness and diversity differ between among connected patches, unconnected patches, and the matrix habitat?

METHODS

Study site

We conducted our study in four of the experimentally fragmented landscapes maintained by the USDA Forest Service at the Savannah River Site (SRS), a National Environmental Research Park in South Carolina, USA (33.208° N, 81.408° W, Figure 1). These landscapes were designed to isolate the effects of patch isolation and connectivitt on plant and animal communities (Tewksbury et al. 2002). Each experimental landscape (i.e., "block") comprises five patches of restored longleaf pine savanna embedded in a matrix of mature pine plantation (Figure 2). The central patch in each block (100 × 100 m) is connected to an identically sized patch by a 150 × 25 m corridor. In addition to this "connected" patch, each block also has "rectangular" and "winged" patches. The rectangular patch is 100 × 137.5 m to account for the area added by the corridor; the winged patch is the same size as the connected patch (100 × 100 m) but includes two 75 × 25 m offshoots to account for the corridor's area and its additional edge. Each block has a duplicate of either the winged or rectangular patch, with all patches 150 m from the central connected patch.

Sampling

In July and August 2024 we sampled dung beetles in four of the SRS blocks (8, 52, 53n, and 54; Figure 1) using pitfall traps. In each block we sampled in the connected patch, a winged patch, a rectangular patch, and in the pine matrix. We placed three traps with openings flush to the ground in the center of each patch; traps were 20 m apart and approximately 40 m from the patch edge. Traps in the matrix were similarly arranged 150 m from the midpoint of the central patch. We sampled each site in each block simultaneously for X days; each sampling date we baited traps between 8-9 pm with ~ ___ g of pig dung wrapped in a coffee filter and then returned 12 hours later to collect the material from the alcohol reservoirs at the base of the trap. Samples were stored in vials of 95% ethanol and returned to the University of Florida for processing. Each block was sampled four times (N = 192 pitfall traps total). All dung beetles were identified to species following Nemes and Price (2015) and Edmonds (2023) and voucher specimens for each species were deposited at the Florida State Collection of Arthropods. We also assigned each species to its foraging strategy (i.e., wallower, tunneller, roller) and identified its habitat preference using previously published studies [CITATIONS].

Biomass estimation

To estimate dung beetle biomass in each patch we bagan by estimating the average biomass of N=15 individuals of the ten most common species we captured. To do so we placed them in a drying oven at — degrees until their collective biomass reached a stable value, then weighed each individual beetle using an Ohaus Adventurer Pro AV53 microbalance. The exception was specimens from three species that were too small to weigh accurately, which from which we estimated average individual biomass by weighing beetles in batches of N = 3 (i.e., *Ateuchus lecontei*), N = 5 (i.e., *Onthophagus pennsylvanicus*), or N = 15 (i.e., *Aphodius alloblackburneus*). The average values for individual biomass were then used to estimate (a) the total biomass of each species in each patch and (b) the total beetle biomass (i.e., all

158 species combined) in each patch.

159 **Species Richness and Diversity**

160 The Richness and Diversity of beetles in each patch was estimated using Hill numbers (Jost
161 2006). Hill numbersd have emerged as the preferred metrics for describing community
162 dynamics and composition for two reasons: first, they are extrapolated from the same
163 equation by manipulating a single parameter (i.e., q) to arrive at estimates of richness and
164 diversity. Second, by manipulating q it is possible to understand compositional shifts in ways
165 that other measures of species richness would obscure (Chao et al. 2014). We compared
166 community composition by increasing magnitudes of diversity components (i.e., qD): *0D* (i.e.,
167 species richness), *1D* (i.e., Shannon entropy), and *2D* (i.e, Simpson Diversity). The values
168 diversity numbers and species richness were calculated using the package `hill` (Li 2018) for
169 the R statistical programming language (Posit team 2025).

170 To test for the effects of connectivity on beetle abundance, species richness, and species
171 diversity we used generalized linear mixed models (i.e., GLMM) fitted to a poisson
172 distribution (Bolker et al. 2009). Patch type was the main effect, with the identity of the
173 sampling block included as a random effect to account for landscape-scale spatial variation.
174 Samples from the different sampling dates were pooled, and because _____, we
175 restricted our analyses of abundance to the six most common species.

176 We used linear mixed-effects models with a Gaussian distribution to determine the
177 effect of patch type on both Shannon Diversity and Simpson's Index (Chao et al. 2014),
178 again including a random effect of block. For both diversity models the block-level random
179 effect standard deviation was slightly greater than the residual error, indicating that
180 variation between blocks accounted for a substantial portion of the overall variability. Finally,
181 we used linear mixed-effects models with a random effect of block to compare (1) total dung
182 beetle biomass and (2) the total biomass of the ten most common species species in different

patch types. We fit all models using the `lme4` package (Bates et al. 2015); the reference level for patch type was the value for the matrix. Prior to conducting our analyses we evaluated the the suitability of our data using *qqplots* generated with the `DARMa` package (Hartig 2024).

RESULTS

We collected a total of N = 5213 beetles: N = 1359 in the Connected patches, N = 1199 in the Winged patches, N = 942 in the Rectangle patches, and N = 1713 in the Matrix. We identified N = 16 species, of which the N = 6 species comprised 93.9% of all captures: *Canthon vigilans* (N = 1473), *Ateuchus lecontei* (N = 1115), *Phanaeus igneus* (N = 958), *Aphodius alloblackburneus* (N = 585), *Dichotomius carolinus* (N = 556), and *Onthophagus pennsylvanicus* (N = 207; Table 1). All but four species were captured in every patch type. *Onthophagus concinnus* was only found in the matrix and winged patches, while *Onthophagus striatulus* was only captured in matrix habitat and rectangular patches. *Geotrupes blackburnii* and *Onthophagus tuberculifrons* were the only species restricted to one patch type (winged and matrix, respectively). All species were within their native ranges.

Dung Beetle Abundance, Richness, & Diversity

There was a significant effect of patch type on average abundance (all species combined), with the matrix and connnected patches having greater abundance than rectangle and winged patches (β = -0.528 and -0.303 respectively, P $<$ 0.001, Table 2). The abundance of the six most abundant species also differed significantly with patch type (Table 6), but the interaction terms indicate these responses were highly idiosyncratic widely (Table 6, Figure 3)

The number of species per patch ranged from N = 8-13 (Figure 6A). There was no significant difference in species richness between connected (β = 0.216, p = 0.839; Table 7),

rectangular (β = -0.548, p = 0.583; Table 7), and winged patches (β = -0.095, p = 0.663; Table 7). **Block effect?**

Biodiversity was also even not significantly different between patch types, however the values of the different metrics varied significantly across sampling blocks (Figure 6B,C). For Shannon Diversity, the estimated mean in matrix patches was 5.202 (SE = 0.665, t = 7.825; Table 8). None of the alternative patch types showed statistically significant differences compared to matrix: connected (β = –0.085, SE = 0.495, t = 0.17), rectangle (β = 0.234, SE = 0.495, t = 0.472), or winged (β = –0.251, SE = 0.495, t = –0.506; Table 8). Results for Simpson's Diversity were similar (Table 9), with the average value in matrix patches was 4.181 (SE = 0.640, t = 6.535). Again, none of the other patch types were significantly different from the others: matrix (β = -0.107, SE = 0.411, t = -0.259), rectangle (β = 0.059, SE = 0.411, t = 0.143), and winged (β = –0.480, SE = 0.411, t = –1.169; Table 9).

Dung Beetle Biomass

Patterns of biomass by patch type were similar to those for abundance: biomass was highest in matrix plots, followed by connected and winged, with biomass in rectangle patches being far lower than in the other locations (Figure 5). There was no significant difference in the total beetle biomass of matrix plots when compared to connected (β = -7.337, SE = 8.659, t = -0.847; Table 3) and winged patches (β = -9.608, SE = 8.659, t = -1.11; Table 3), but biomass in rectangle patches was significantly different (β = -23.296, SE = 8.659, t = -2.69; Table 3). Block effects continue to be important for explaining high variation as the total biomass collected in block 52 was nearly quadruple that collected in Block 8 (Figure 5). Linear mixed effects models also indicate significant differences between the biomass of different species (Table 4, Figure 4).

DISCUSSION

This study advances our understanding of the factors shaping dung beetle community composition in temperate regions of the southeastern United States. In addition, the experimental design enables direct comparisons between populations in continuous matrix habitat and those in both isolated and corridor connected patches. Our main findings emphasized: (1) Habitat type and patch shape were the main driving factors for determining how dung beetle species abundances were composed, however effects were species specific. (2) Patch shape and isolation had less of an influence on species richness which was relatively even on both a patch and block level. (3) Species diversity metrics were also relatively even across patch types however varied widely by sampling blocks. These results suggest that while there may indeed be effects of patch structure and connectivity on dung beetle abundances and community composition, other landscape scale drivers appear to be more prominent for species richness and diversity.

In our comparison of dung beetle compositions, connectivity and habitat edge accounted for differences in dung beetle abundances, yet species richness and diversity were even across patch types and in the matrix. Total beetle counts were consistently highest in the matrix, in comparison abundances is connected patches were not significantly different while those of the winged and rectangular patches were lower. Additionally, we also observed that patch effects were not equally proportional for all species. This suggests that the species captured could be exhibiting habitat preferences between the open patches versus forested matrix. Another potential explanation is that species in the matrix are acting as a source population which feeds into patches with edge acting as a drift fence for directing movement. The latter example seems more likely - small mammals at SRS, which are a potential source of dung, appear to preferentially use matrix habitat (Mabry et al. 2003) and flies, which have similar resource dependencies were also found to have a similar interaction (Fried et al. 2005). Moreover studies on dung beetles in the tropics found more dramatic differences in

dung beetle populations in fragments and matrix (Barragan et al. 2011).

Although richness and diversity were the same among treatments, there was notable variability between sampling blocks. Block 8 generally had the lowest species richness and biodiversity while 53n had the highest. While the experimental design attempted to control for the effects of patch size and edge, there could be large (and potential unknown) environmental gradients across the SRS landscape that could influence the observed patterns in diversity and abundance. For instance, at the time of our study Block 8 had the densest matrix of any of the blocks. This could have hindered the diffusion of bait scent, leading to lower capture rates in this block. Other habitat characteristics that might differ among blocks could have been influential as well – for example, soil quality and forest cover can determine where beetles can reproduce (Arellano et al. 2008, Conover et al. 2019). The same is true for land-use history; much of the SES land would have previously been used for agriculture, and during development of experimental units cleared with heavy machinery might experience heavy soil compaction *brudvig*. Landscape features could also affect mammal movement, which might in turn limit dung availability - Block 8 was nearest to roadways used by all employees at SRS, and the eastern side of the site had large bodies of water upon which mammals are dependent (Harvey et al. 2006, Dechen Quinn et al. 2013, Barahona-Segovia 2021). Additionally, deer and feral hogs are the most dominant mammals at the site, two species that do well in patchy landscapes (Castillo-Contreras et al. 2018, Fraser et al. 2019). The locally variable patterns of abundance in the landscape of these generalist large mammals, coupled with ability to easily move through modified landscapes, might help explain the limited variation in richness or diversity within blocks (i.e., between patch types) but large block effects. Deer also commonly forage along the edges of habitat patches and forest ecotones. If they are spending more time in these locations, the higher abundance and diversity in connected and winged patches might in part be due to 'drift-fence' effects **citation of levey drift fence**.

Species-specific differences are apparent but do not follow any particular trend. *Aphodius alloblackburneus* had a disproportionate positive effect to being captured in matrix patches as compared to other species. Beetles from the genus *Aphodius* do tend to show patterns of habitat specificity (Roslin and Koivunen 2001), and many preferring forested habitat (Frank et al. 2017b), so it is not unexpected that they might show a preference towards forested matrix. We did not detect that any species was more positively associated with open patches despite suggestions that some species (e.g., *Canthon vigilans*, *Melanocanthon bispinatus*) prefered open habitat (Nealis 1977, Conover et al. 2019). This could be another sign pointing towards matrix acting as a source population, and since open habitat was much less dominant in our experimental system beetles could be moving into patches from habitat edge.

The patterns in dung beetle biomass largely echo what we observed for abundance. Although biomass is understood to be positively associated with dung removal (Slade et al. 2011), dung beetle species vary greatly in terms of morphology and functionality (Ospina-Garcés et al. 2018) so evaluating species-specific patterns is particularly important. Because of this we can expect that the magnitude of removal might be greater in areas of larger biomass (i.e. matrix or connected patches), however future work should aim to directly measure this potential pattern. Indeed comparing the biomass of different species emphasizes the morphological variability between our study species - some of the most common species (e.g., *Ateuchus lecontei*, *Aphodius allobloackburneus*) contribute very little to total biomass (Figure 4). In addition to this, dung beetles appear to be an exception to the globally and taxonomically robust rule that abundance is negatively correlated with individual biomass (White et al. 2007). More intensive sampling could determine whether this trend is truly apparent.

Despite ample work documenting their ecological importance, there is a surprising lack of research on dung beetle diversity and corridors outside of the tropics (Nichols et al. 2007),

with work in temperate locations coming primarily from Europe where *Aphodius* beetles dominate (Roslin and Koivunen 2001). Many studies conducted in the tropics found strong preference by some species for open fields vs. continuous forest (Damborsky et al. 2015), while othersfound that forest species were using living fences as corridors (Arellano et al. 2008). We did not observe such intense habitat specificity, which may not be especially common outside of the tropics. We found similar patterns of community composition as did the small number of prior dung beetle biodiversity surveys conducted in the southeastern United States (Nealis 1977, Conover et al. 2019), including at least one more where differences between treatment types were non-significant while variation on a larger scale was more apparent (Young et al. 2023). We did expect some species to be more dominant in patches with higher edge ratios as was found with ants at SRS (Resasco et al. 2014), but species were evenly present in all patches.

Finally, it is important to emphasize that while dung beetles are capable of long distance flight and detecting dung at distances of over 50 meters (Gray et al. 2022), the results we observed are likely the result of a mismatch between the spatial scale of the experimental replicates and dung beetle movement. Put another way, some beetles were almost certainly drawn by the dung used in baits from the matrix into the plots. If this were an overarching effect we'd expect all plots to be similar to the matrix, or at least to each other. The fact that rectangular are have lower abundance suggests that some sort of landscape effect is apparent, likely related to habitat edge since connected and winged patches were the most similar. To remedy this issue any future work in this site should either focus solely on dung beetles with more limited dispersal ability or consider conducting mark-release-recapture experiments in an attempt to document movements within and between patches.

There were some limitations of this study which should be addressed in any future work to capture a clearer picture of how populations are being altered by habitat connectivity.

First, sampling for this study was conducted over a two month period in the summer of 2024, sufficient data was collected but due to a lack of available resources temporal patterns were obscured due to inconsistent sampling periods. This is highly important since dung beetles exhibit consistent patterns of seasonality (Davis 1966, Conover et al. 2019). Another potential avenue for improvement is lowering the grain size of sampling and changing trap placement to better understand the effects of edge proximity and connectivity. We also used one of the best bait types for collecting dung beetles in pig dung (Marsh et al. 2013), but a mix of differently sourced baits may have been more optimal as more diverse baits attract more diverse species (Frank et al. 2017a, 2018, Giménez Gómez et al. 2021).

Moving forward, there is a bounty of knowledge yet to be collected for dung beetles in the south eastern United States. Relative to what is already known in tropical ecosystems, this study provides a glimpse on how dung beetles are responding to fragmentation and connectivity in subtropic pine habitats, but much can still be learned about dung beetles in these spaces and the role they play in ecosystems as well as how they interact with landscapes. The unique role of dung beetles as waste removers is fascinating enough and of great importance for maintaining landscapes so dominated by pastures like Southern US (Slade et al. 2016a, Cheng et al. 2022). Not only waste removal but also secondary effect such as seed dispersal and parasitic reduction are important to understand in the context of fragmentation (Fincher 1975, Vulinec 2002). SRS provides an excellent experimental design for direct comparison of landscapes but another main goal of the site is to study dispersal, dung beetles would be an excellent system for studying movement. They already show promise, in preliminary trials we released beetles from both the connected and rectangular patches with baited traps in the central patch. We recovered one recapture which originated from the connected patch, indicating that individuals do move through corridors.

Although there is plentiful work to be done, this study paints an encouraging picture that while dung beetles face disturbances that make habitats less preferential, they are

robust enough to persist throughout fragmented landscapes and the management strategies we do have to connect fragmented landscapes work to mitigate losses. Because assemblages remain mostly the same across these landscapes, ecosystem services should remain uninterrupted allowing continuing benefits to all community members. While this study of dung beetles in fragmented landscapes was conducted in a relatively protected area, in real world application we could consider the use of movement corridors within disturbed areas to help bolster effected populations, but also near areas of high importance such as pastures to provide areas of refuge for beetles. In this case it is important to consider the inverse of what was manipulated in this study where open field acts as matrix and forested area is focal landscape, thus it is vital to continue learning more about how landscape composition and connectivity effect dung beetle populations.

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Table 1
Dung beetle species sampled in the SRS site and their total abundance over the course of the study.

Species	**Guild**	**N**	**Matrix**	**Corridor**	**Winged**	**Rectangular**
Canthon vigilans	roll	1473	x	x	x	x
Ateuchus lecontei	tunnell	1115	x	x	x	x
Phanaeus igneus	tunnell	958	x	x	x	x
Dichotomius carolinus	tunnell	556	x	x	x	x
Aphodius alloblackburneus	dwell	585	x	x	x	x
Onthophagus pennsylvanicus	tunnell	207	x	x	x	x
Melanocanthon bispinatus	roll	83	x	x	x	x
Phanaeus vindex	tunnell	133	x	x	x	x
Boreocanthon probus	roll	47	x	x	x	x
Copris minutus	tunnell	24	x	x	x	x
Deltochilum gibbosum	roll	14	x	x	x	x
Aphodius oximus	dwell	11	x	x	x	x
Geotrupes blackburnii	tunnell	1			x	
Onthophagus concinnus	tunnell	2	x		x	
Onthophagus striatulus	tunnell	3	x			x
Onthophagus tuberculifrons	tunnell	1	x			

Table 2
Model of species abundance by patch type.

Effect	Group	Term	Estimate	SE	Statistic	P
Fixed		(Intercept)	5.896	0.294	20.062	< 0.001
Fixed		Patch type: Connected	-0.125	0.139	-0.9	0.368
Fixed		Patch type: Rectangle	-0.528	0.107	-4.919	< 0.001
Fixed		Patch type: Winged	-0.303	0.072	-4.188	< 0.001
Random	Block	SD (Intercept)	0.585			
Random	Block	correlation: (Intercept) x Patch type: Connected	-0.932			
Random	Block	correlation: (Intercept) x Patch type: Rectangle	-0.744			
Random	Block	correlation: (Intercept) x Patch type: Winged	-0.967			
Random	Block	SD Patch type: Connected	0.266			
Random	Block	correlation: Patch type: Connected x Patch type: Rectangle	0.865			
Random	Block	correlation: Patch type: Connected x Patch type: Winged	0.816			
Random	Block	SD Patch type: Rectangle	0.194			
Random	Block	correlation: Patch type: Rectangle x Patch type: Winged	0.653			
Random	Block	SD Patch type: Winged	0.116			

Table 3
Linear mixed effects model of pooled total biomass by patch type including sampling block as a random effect.

Effect	Group	Term	Estimate	SE	Statistic
Fixed		(Intercept)	61.593	17.199	3.581
Fixed		Patch: Connected	-7.337	8.659	-0.847
Fixed		Patch: Rectangle	-23.296	8.659	-2.69
Fixed		Patch: Winged	-9.608	8.659	-1.11
Random	block	SD (Intercept)	32.145		
Random	Residual	SD (Observation)	12.246		

Table 4
Linear mixed effects model of biomass by species and patch type including sampling block as a random effect. Species codes = bpro: Boreocanthon probus, cvig: Canthon vigilans, dcar: Dichotomius carolinus, mbis: Melanocanthon bispinatus, ocon: Onthophagus concinnus, open: Onthophagus pennsylvanicus, pign: Phanaeus igneus, pvin: Phanaeus vindex

Effect	**Group**	**Term**	**Estimate**	**SE**	**Statistic**
Fixed		(Intercept)	1.047	2.226	0.471
Fixed		Species: alec	0.572	1.949	0.293
Fixed		Species: bpro	0.078	1.949	0.04
Fixed		Species: bvig	13.639	1.949	7
Fixed		Species: dcar	20.454	1.949	10.497
Fixed		Species: mbis	0.090	1.949	0.046
Fixed		Species: ocon	-0.040	1.949	-0.02
Fixed		Species: open	-0.012	1.949	-0.006
Fixed		Species: pign	14.484	1.949	7.433
Fixed		Species: pvin	1.853	1.949	0.951
Fixed		Patch: Connected	-0.734	1.232	-0.595
Fixed		Patch: Rectangle	-2.330	1.232	-1.89
Fixed		Patch: Winged	-0.961	1.232	-0.78
Random	block	SD (Intercept)	3.154		
Random	Residual	SD (Observation)	5.511		

Table 5
Results of an analysis of assesing the effects of species identity, patch type, and their interaction on the abundance of dung beetle species.

Term	**Statistic**	**df**	**P**
Species	1044.680	5	< 0.001
Patch type	254.638	3	< 0.001
Species x Patch type	110.372	15	< 0.001

Table 6
Generalized linear mixed model of abundance by patch type of the 6 most abundant species and interaction term for patch type and species including sampling block as a random effect.

Effect	**Group**	**Term**	**Estimate**	**SE**	**Statistic**	**P**
Fixed		(Intercept)	4.17	0.261	15.948	< 0.001
Fixed		Species: alec	0.234	0.078	2.989	0.003
Fixed		Species: cvig	0.457	0.075	6.107	< 0.001
Fixed		Species: dcar	-0.518	0.096	-5.407	< 0.001
Fixed		Species: open	-1.373	0.13	-10.549	< 0.001
Fixed		Species: pign	-0.105	0.085	-1.231	0.218
Fixed		Patch type: Connected	-0.749	0.103	-7.257	< 0.001
Fixed		Patch type: Rectangle	-1.487	0.136	-10.913	< 0.001
Fixed		Patch type: Winged	-1.188	0.121	-9.815	< 0.001
Fixed		Species: alec x Patch type: Connected	0.495	0.13	3.809	< 0.001
Fixed		Species: cvig x Patch type: Connected	0.595	0.124	4.8	< 0.001
Fixed		Species: dcar x Patch type: Connected	0.567	0.153	3.715	< 0.001
Fixed		Species: open x Patch type: Connected	0.573	0.201	2.852	0.004
Fixed		Species: pign x Patch type: Connected	0.761	0.135	5.636	< 0.001
Fixed		Species: alec x Patch type: Rectangle	0.899	0.162	5.561	< 0.001
Fixed		Species: cvig x Patch type: Rectangle	0.834	0.158	5.287	< 0.001
Fixed		Species: dcar x Patch type: Rectangle	0.828	0.188	4.399	< 0.001
Fixed		Species: open x Patch type: Rectangle	0.794	0.243	3.266	0.001
Fixed		Species: pign x Patch type: Rectangle	1.228	0.165	7.434	< 0.001
Fixed		Species: alec x Patch type: Winged	0.819	0.146	5.611	< 0.001
Fixed		Species: cvig x Patch type: Winged	0.987	0.14	7.072	< 0.001
Fixed		Species: dcar x Patch type: Winged	1.02	0.165	6.182	< 0.001
Fixed		Species: open x Patch type: Winged	0.41	0.24	1.711	0.087
Fixed		Species: pign x Patch type: Winged	1.037	0.151	6.853	< 0.001
Random	Block	SD (Intercept)	0.509			

Table 7
Linear mixed effects model of species richness by patch type.

Effect	**Term**	**Estimate**	**SE**	**Statistic**	**P**
Fixed	(Intercept)	2.398	0.151	15.906	< 0.001
Fixed	Patch Type: Connected	-0.047	0.216	-0.216	0.829
Fixed	Patch Type: Rectangle	-0.121	0.22	-0.548	0.583
Fixed	Patch Type: Winged	-0.095	0.218	-0.436	0.663

Table 8
Linear mixed effects model of Shannon Diversity by patch type.

Effect	**Group**	**Term**	**Estimate**	**SE**	**Statistic**
Fixed		(Intercept)	5.202	0.665	7.825
Fixed		Patch type: Connected	0.085	0.495	0.171
Fixed		Patch type: Rectangle	0.234	0.495	0.472
Fixed		Patch type: Winged	-0.251	0.495	-0.506
Random	Block	sd__(Intercept)	1.13		
Random	Residual	SD (Observation)	0.701		

Table 9
Linear mixed effects model of Simpson's Diversity by patch type.

Effect	**Group**	**Term**	**Estimate**	**SE**	**Statistic**
Fixed		(Intercept)	4.181	0.64	6.535
Fixed		Patch type: Connected	-0.107	0.411	-0.259
Fixed		Patch type: Rectangle	0.059	0.411	0.143
Fixed		Patch type: Winged	-0.48	0.411	-1.169
Random	Block	sd__(Intercept)	1.14		
Random	Residual	sd__Observation	0.581		

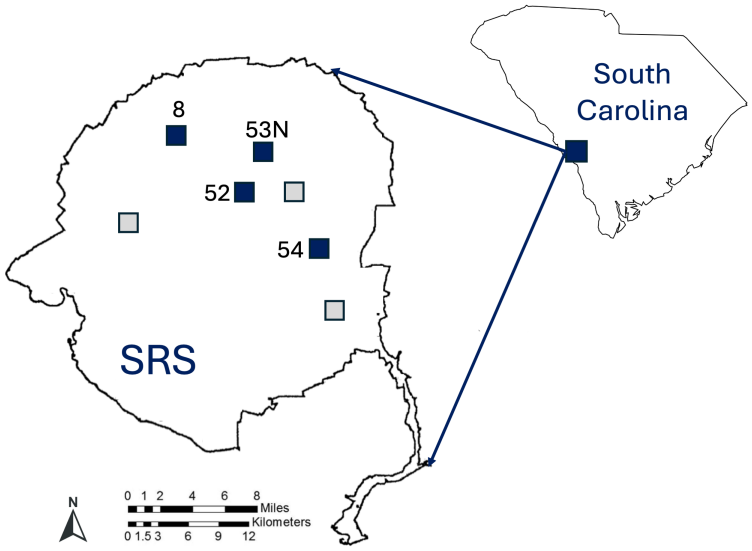


Figure 1. All experimental blocks at SRS. Filled squares indicate the blocks in which dung beetles were sampled. Each block has a unique ID number.

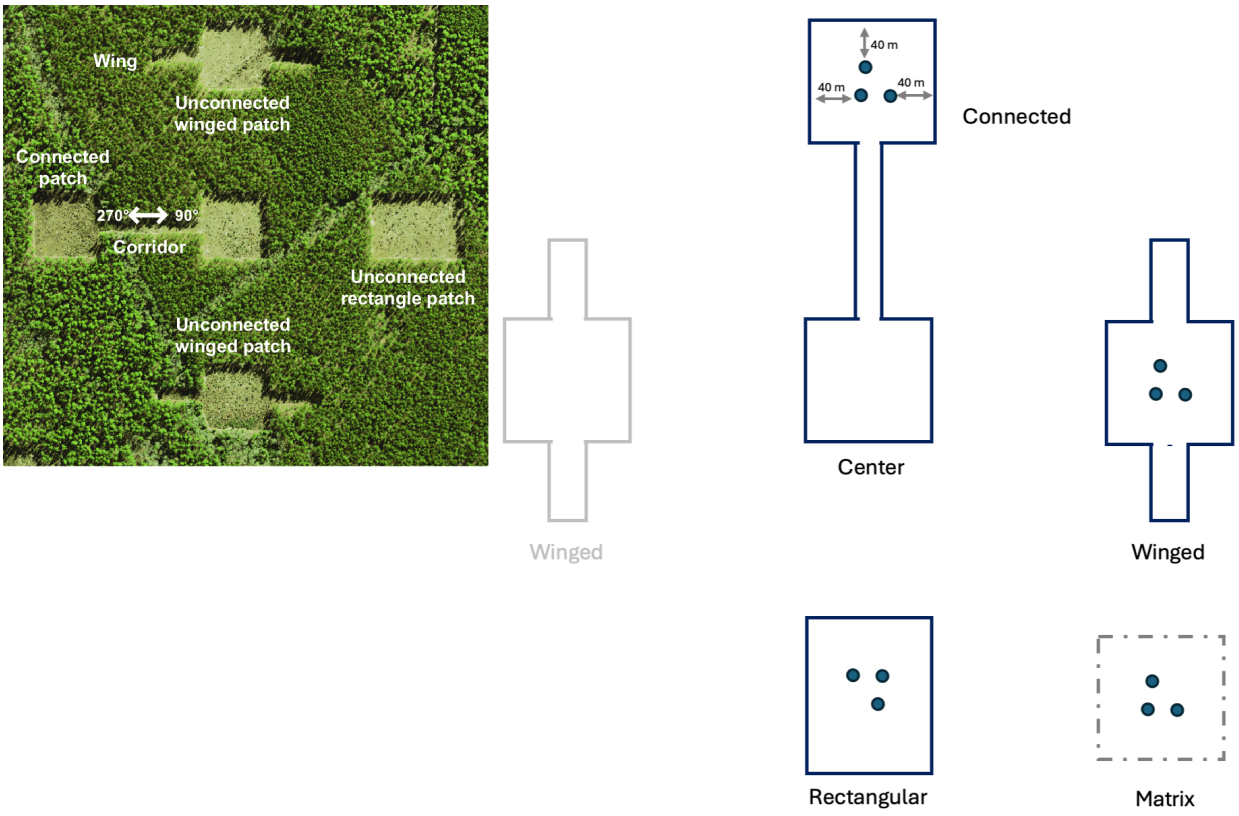


Figure 2. Experiemental patches at the SRS

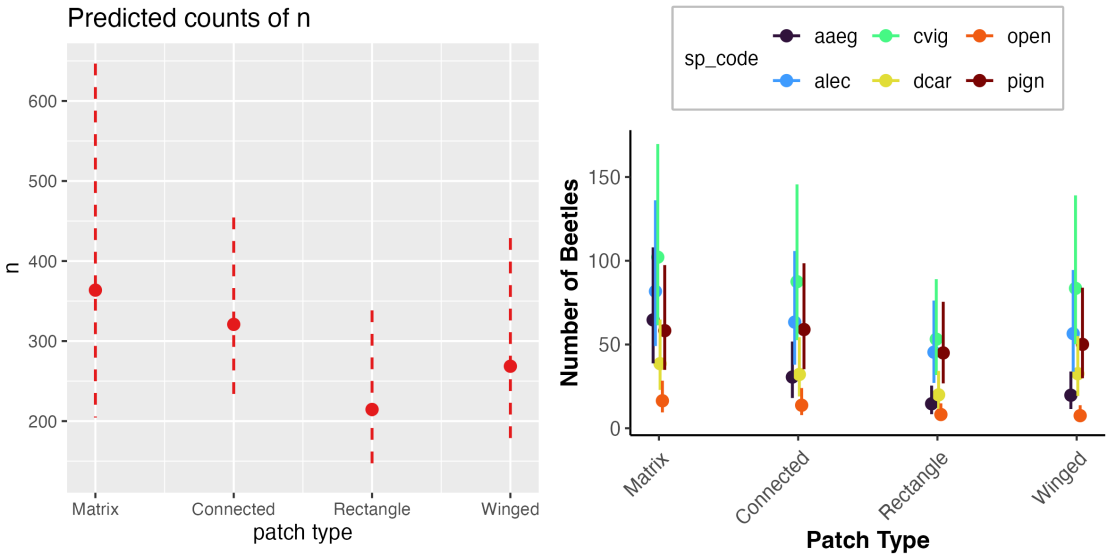


Figure 3. Mean abundance in each SRS patch type of (A) all dung beetle species combined and (B) each of the six most abundant dung beetle species. Species codes: alec: Ateuchus lecontei, cvig: Canthon vigilans, dcar: Dichotomius carolinus, open: Onthophagus pennsylvanicus, pign: Phanaeus igneus, aaeg: Aphodius alloblackburneus.

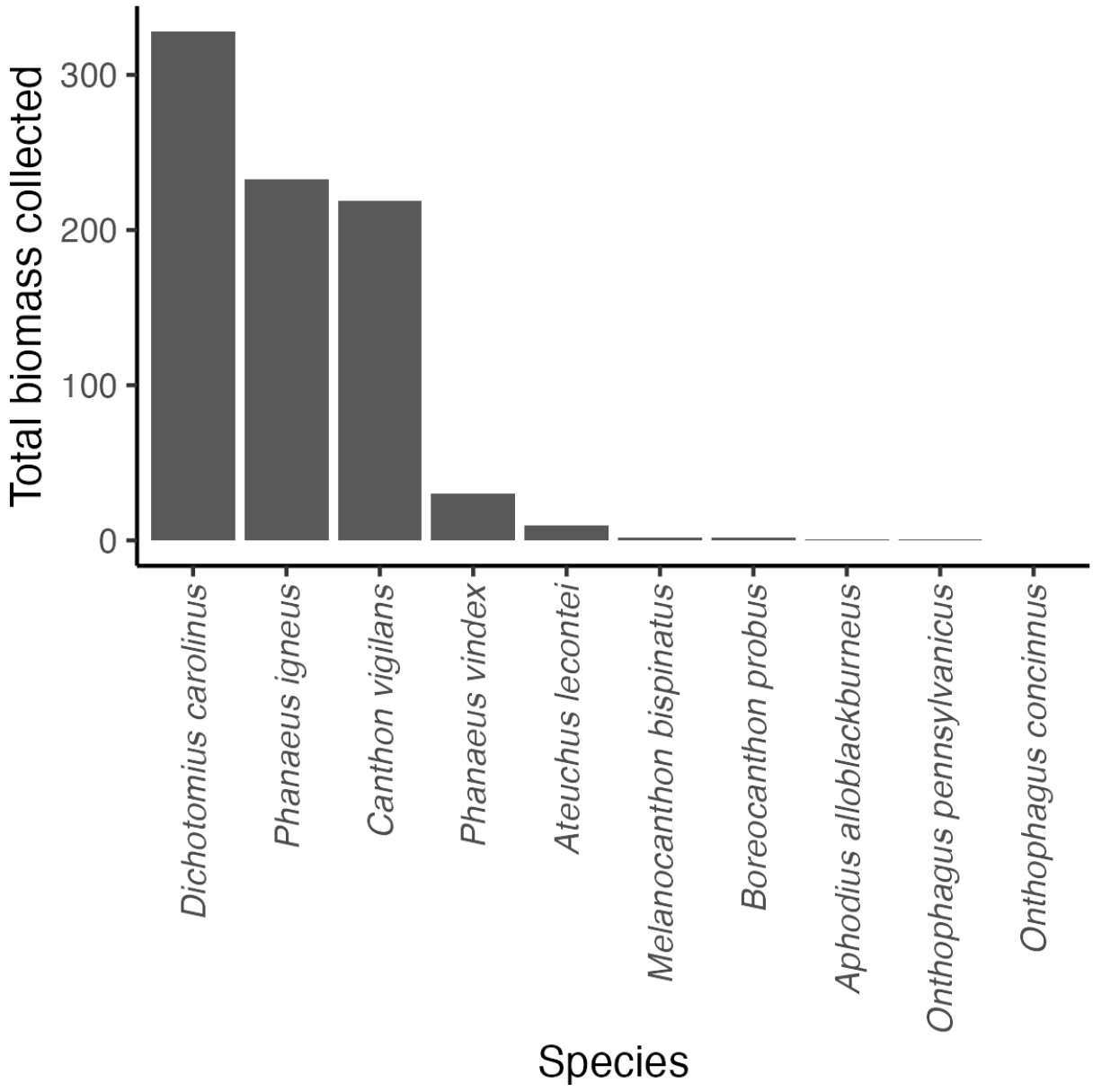


Figure 4. Total biomass of the ten most common dung beetle species captured at the SRS.

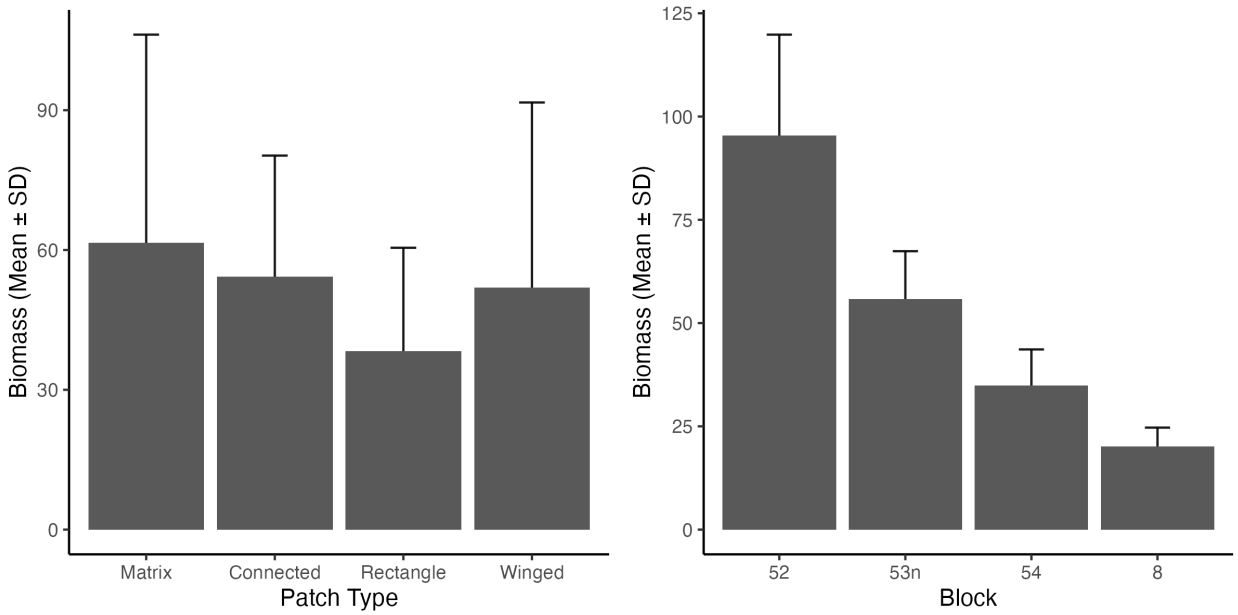


Figure 5. Average total biomass of beetles by (A) patch type and (B) block.

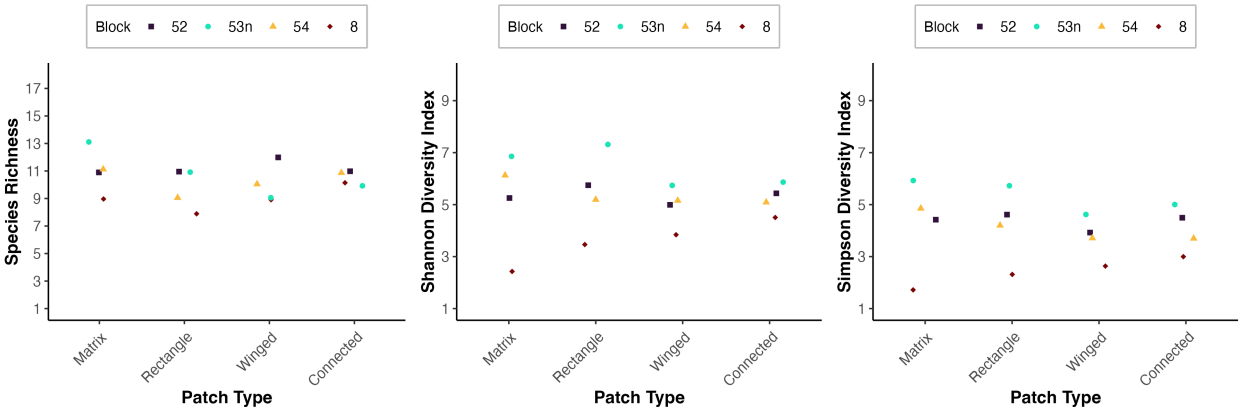


Figure 6. (A) Species richness, (B) Shannon Diversity, and (C) Simpson's Diversity of dung beetle species in each of the SRS patch types. The shape of the points indicates the block in which a patch was located.